

Software Requirements Specification (SRS)

CocoConnect – A Digital Ecosystem for Coconut Estate Management

Version: 3

Prepared by: Group 38

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements for **CocoConnect**, a digital platform developed to improve the management of coconut estates and connect stakeholders within the coconut industry. The document serves as a reference for developers, testers, project supervisors, and stakeholders throughout the software development life cycle.

CocoConnect aims to solve several challenges currently faced by coconut estate owners, including difficulties in selling products, finding reliable workers and service providers, maintaining harvest records, and establishing trust between buyers and sellers. The system provides an integrated platform where stakeholders can manage estates, trade coconut products, hire workers, record harvest information, and build trusted business relationships through a rating and review system.

This document provides a detailed description of the system requirements to ensure that the final product meets stakeholder expectations and business objectives.

1.2 Scope

CocoConnect is a cross-platform application designed to digitalize the operations of the coconut industry by bringing together product trading, workforce management, estate management, and harvest monitoring within a single platform.

The system allows estate owners and sellers to advertise coconut products while enabling buyers to search for products based on their requirements. It also provides a workforce

marketplace where estate owners can find skilled workers and service providers such as coconut pluckers, huskers, fertilizer workers, tractor suppliers, and equipment suppliers.

In addition, the system supports harvest recording, estate management, and a rating and review mechanism to improve trust among users. The platform is designed with future scalability in mind, allowing additional agricultural products such as pepper, mango, lemon, orange, pineapple, and poultry products to be included in later versions.

The first version of CocoConnect focuses on supporting the coconut industry while providing a flexible architecture for future expansion.

1.3 Intended Audience

This document is intended for the following stakeholders:

- Project Supervisors
- Industry Client and Stakeholders
- Software Developers
- Business Analysts
- UI/UX Designers
- Quality Assurance Engineers
- Database Developers
- System Testers
- Future Maintenance Team Members

Each stakeholder can use this document to understand the system requirements, project scope, and expected functionality before and during development.

1.4 Definitions, Acronyms and Abbreviations

Term	Description
SRS	Software Requirements Specification
BA	Business Analyst
SE	Software Engineer
QA	Quality Assurance
UI	User Interface
UX	User Experience
API	Application Programming Interface

DBMS	Database Management System
Admin	Platform Administrator
Estate Manager	Person responsible for managing daily estate operations and recording harvest information
Service Provider/Worker	A user who offers services such as coconut plucking, husking, fertilizing, harvesting, tractor supply or equipment rental,
Product Marketplace	Module used to buy and sell coconut products
Workforce Marketplace	Module used to hire workers and service providers

1.5 References

The following resources were used during requirement gathering and system analysis:

- Industry Project Proposal – CocoConnect
- Stakeholder interviews conducted with the Coconut Growers Association of Sri Lanka
- Estate visit observations and requirement gathering sessions
- Existing system analysis (HelpGrow and Polkaru)
- IEEE Recommended Practice for Software Requirements Specifications (IEEE 29148)

1.6 Document Overview

This document is organized into seven main sections.

- **Section 1** introduces the purpose, scope, audience, terminology, and references related to the project.
- **Section 2** provides an overview of the proposed system, including product functions, user categories, operating environment, assumptions, and constraints.
- **Section 3** specifies the functional requirements of each module, including user management, product marketplace, workforce marketplace, estate management, harvest management, rating system, notifications, and administration.
- **Section 4** describes the external interface requirements, including user interfaces, software interfaces, hardware interfaces, and communication interfaces.
- **Section 5** outlines the non-functional requirements such as performance, security, reliability, usability, scalability, and maintainability.
- **Section 6** presents the business rules that govern system operations.

- **Section 7** describes future enhancements planned for upcoming versions of CocoConnect.

2. Overall Description

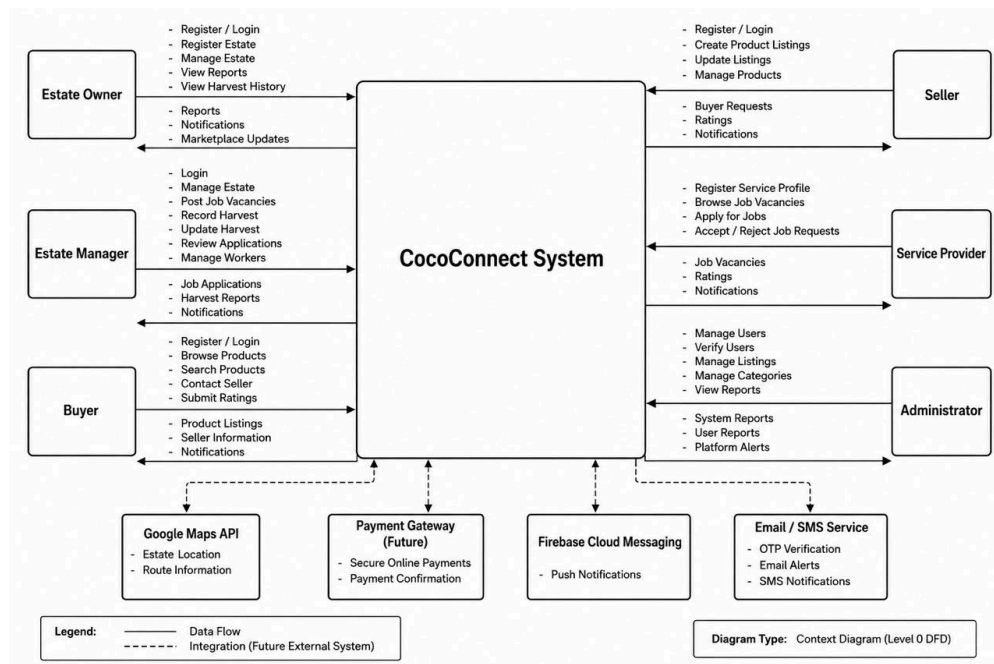
2.1 Product Perspective

CocoConnect is a new web and mobile-based platform designed to support the digital transformation of the Sri Lankan coconut industry. The system combines multiple services into a single ecosystem, reducing the need for separate applications or manual processes.

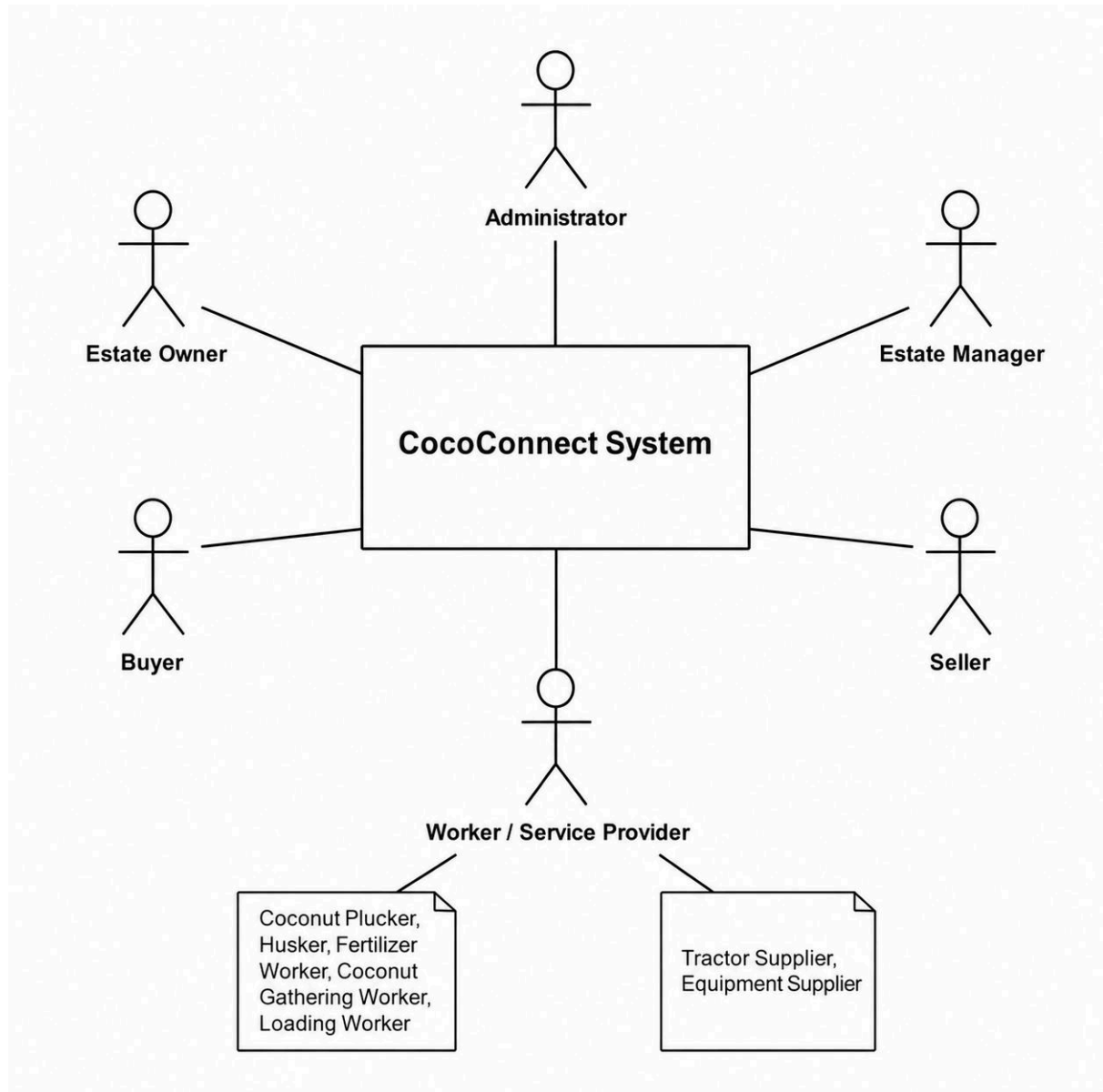
The platform consists of several integrated modules, including User Management, Product Marketplace, Workforce Marketplace, Estate Management, Harvest Management, Rating & Review System, Notification Management, and Administration.

The application will be developed using Flutter for the frontend, Node.js for backend services, and PostgreSQL for database management.

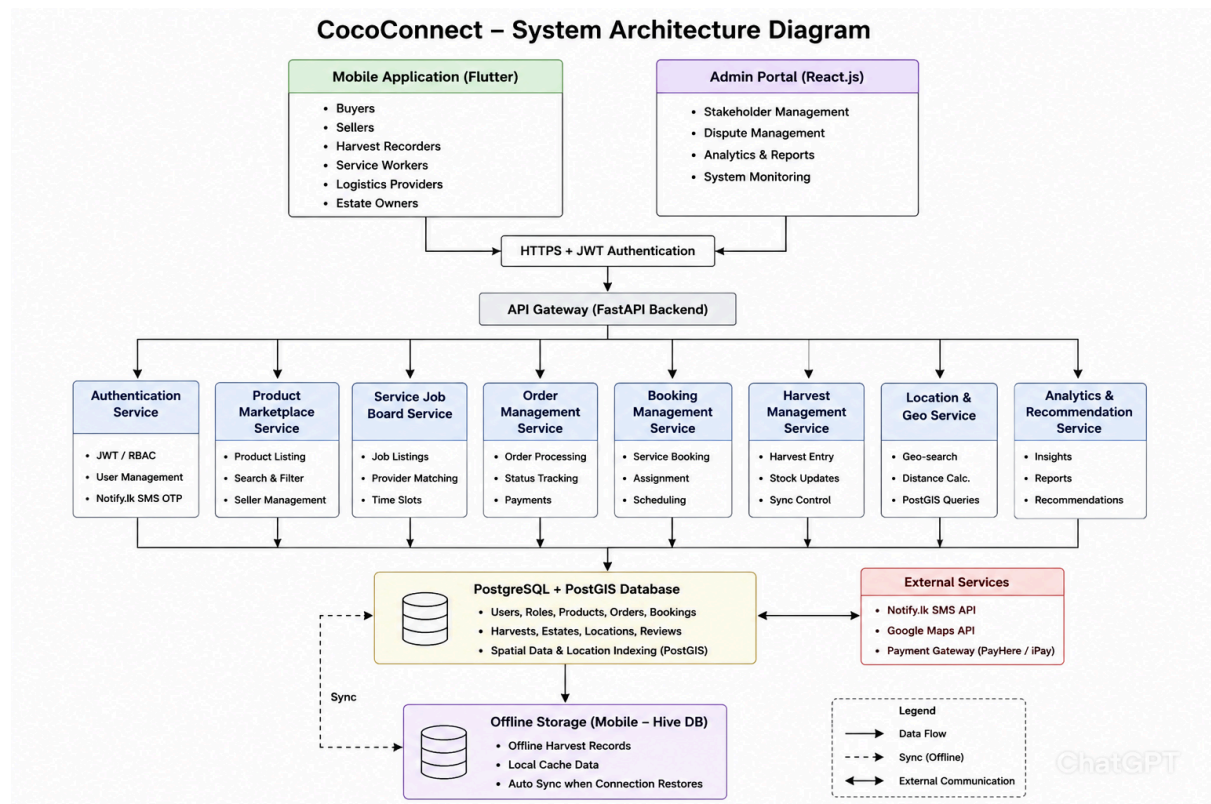
2.1.1 System Context Diagram



2.1.2 Stakeholder Diagram



2.1.3 System Architecture Diagram



2.2 Product Functions

The major functions of CocoConnect include:

- User registration and authentication
- Profile management
- Estate registration and management
- Product listing and marketplace management
- Product browsing and searching
- Workforce and service provider hiring
- Job request management
- Harvest recording and history management
- Rating and review management
- Notification management
- Administrative user verification and platform monitoring
- Reporting and analytics

2.3 User Classes and Characteristics

Estate Owner

Owns one or more coconut estates and oversees estate operations. The Estate Owner registers estates, monitors harvest performance, views reports, and manages product listings (if acting as a seller).

Estate Manager

Manages the day-to-day operations of an assigned estate. The Estate Manager posts job vacancies, reviews job applications, coordinates workers and service providers, records harvest information, and updates estate records on behalf of the Estate Owner.

Buyer

Searches for coconut products, views seller information, and contacts sellers to purchase products.

Seller

Lists coconut products for sale, manages product availability, and communicates with buyers.

Worker/Service Provider

Provides labour services such as coconut plucking, husking, fertilizing, harvesting, loading, offers machinery or equipment-related services such as tractor services or equipment rental.

Administrator

Manages users, verifies accounts, monitors listings, resolves disputes, and maintains platform integrity.

2.4 Operating Environment

The system will operate on:

- Android mobile devices
- iOS mobile devices
- Modern web browsers (Google Chrome, Microsoft Edge, Mozilla Firefox)

Software Environment:

- Flutter

- Node.js
- Express.js
- PostgreSQL
- REST API

2.5 Design Constraints

- Users require an internet connection for most platform features.
- The system must support role-based access control.
- Sensitive user information must be securely stored.
- The application should provide a responsive interface across mobile and web platforms.
- The system should be designed to support future feature expansion without major architectural changes.

2.6 Assumptions and Dependencies

Assumptions

- Users possess basic digital literacy.
- Users have access to smartphones or computers with internet connectivity.
- Stakeholders provide accurate information during registration and product listing.
- Workers and service providers maintain updated profiles.

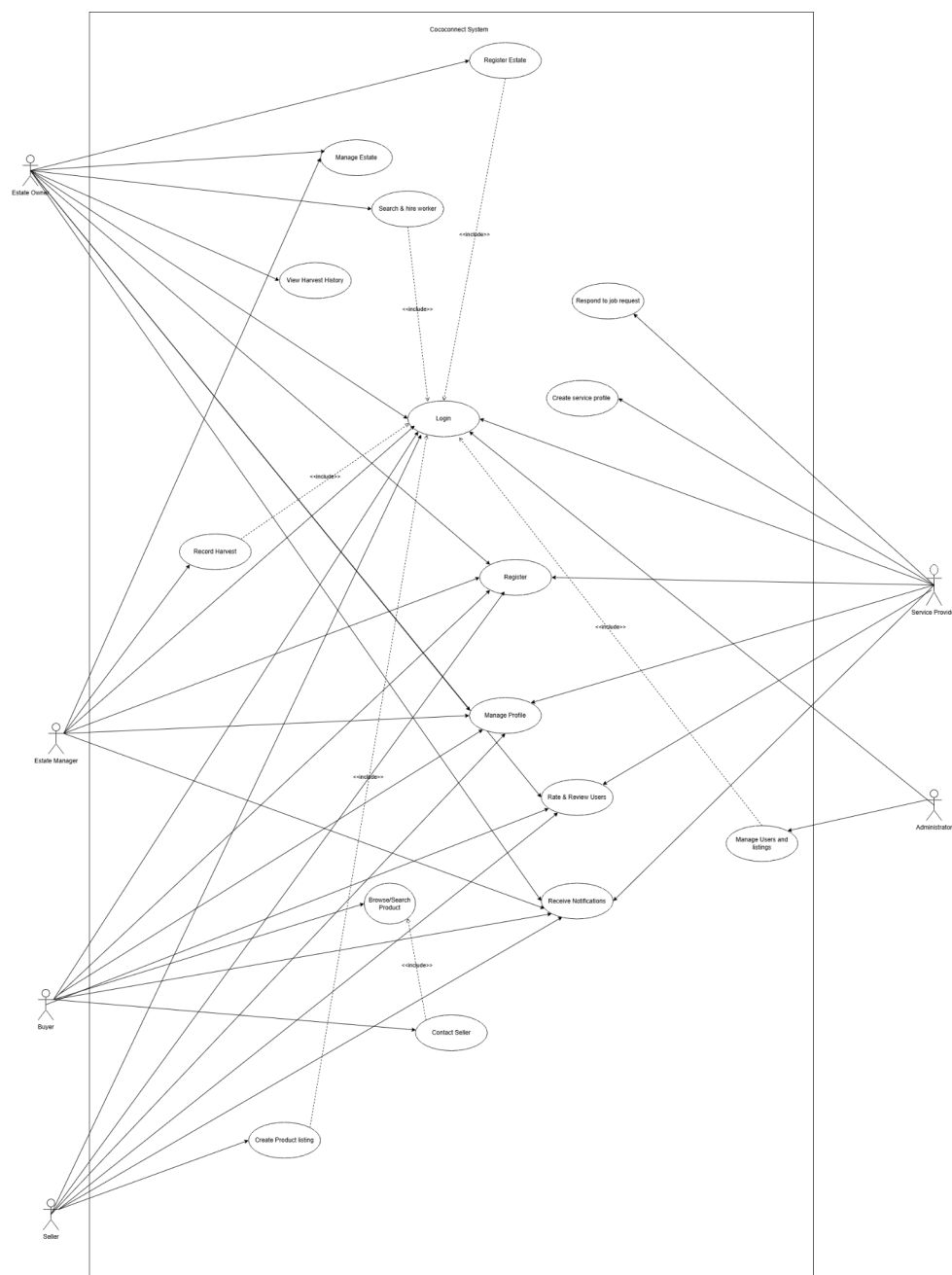
Dependencies

- Stable internet connectivity.
- Availability of the PostgreSQL database server.
- Third-party notification services.
- Future integration with online payment gateways if implemented.

3. System Features (Functional Requirements)

Overall Use Case Model

The Overall Use Case Diagram provides a high-level view of the CocoConnect system, illustrating the primary actors and the major functionalities they interact with. It serves as an overview before presenting the detailed use case diagrams for each functional module.



High-Level Description Table

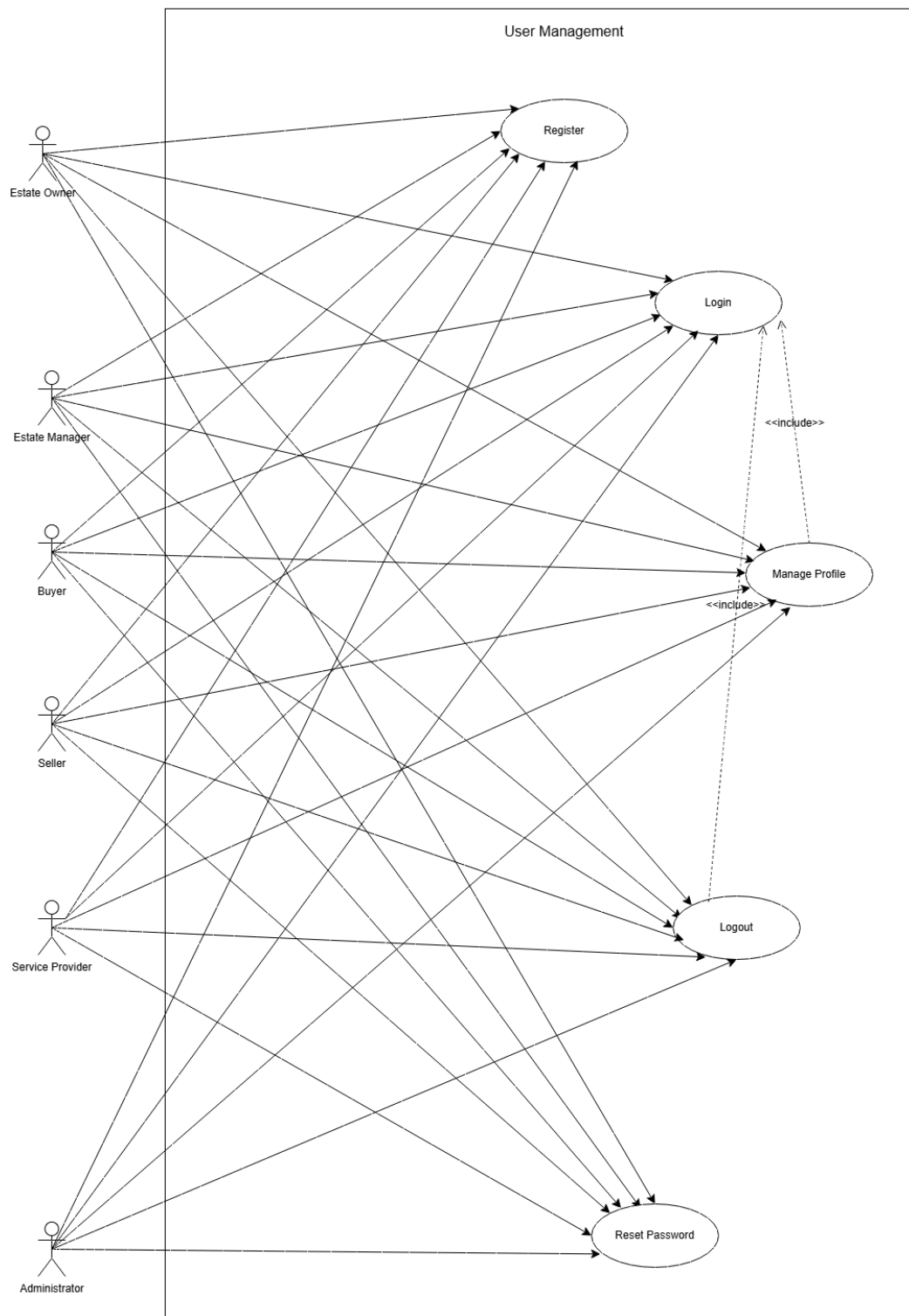
Item	Description
Diagram Name	Overall Use Case Diagram
Purpose	Illustrates the overall functionality of the CocoConnect system and the interactions between all actors and the system.
Primary Actors	Estate Owner, Estate Manager, Buyer, Seller, Service Provider, Administrator
Related Modules	User Management, Product Marketplace, Workforce Marketplace, Estate & Harvest Management, Rating & Notification, Administration
Precondition	Users access the system according to their assigned roles.
Postcondition	Users perform the required system functions based on their permissions.

3.1 User Management Module

Description

The User Management module allows users to register, log in, manage their profiles, and access system features based on their assigned roles. It ensures secure authentication and role-based authorization for all users.

3.1.1 User Management Use Case Diagram



High-Level Description

Item	Description
Diagram Name	User Management
Purpose	Describes how users register, log in, manage their profiles, and log out of the system.
Primary Actors	Estate Owner, Estate Manager, Buyer, Seller, Service Provider, Administrator
Related Functional Requirements	FR-01 User Registration, FR-02 Login, FR-03 Profile Management
Precondition	A user accesses the CocoConnect platform.
Postcondition	User account is created or the user successfully accesses their dashboard.

Functional Requirements

FR-01: The system shall allow users to register by providing personal information and selecting a user role.

FR-02: The system shall allow registered users to log in using their email address and password.

FR-03: The system shall allow users to reset forgotten passwords.

FR-04: The system shall allow users to update their profile information.

FR-05: The system shall allow users to upload a profile picture.

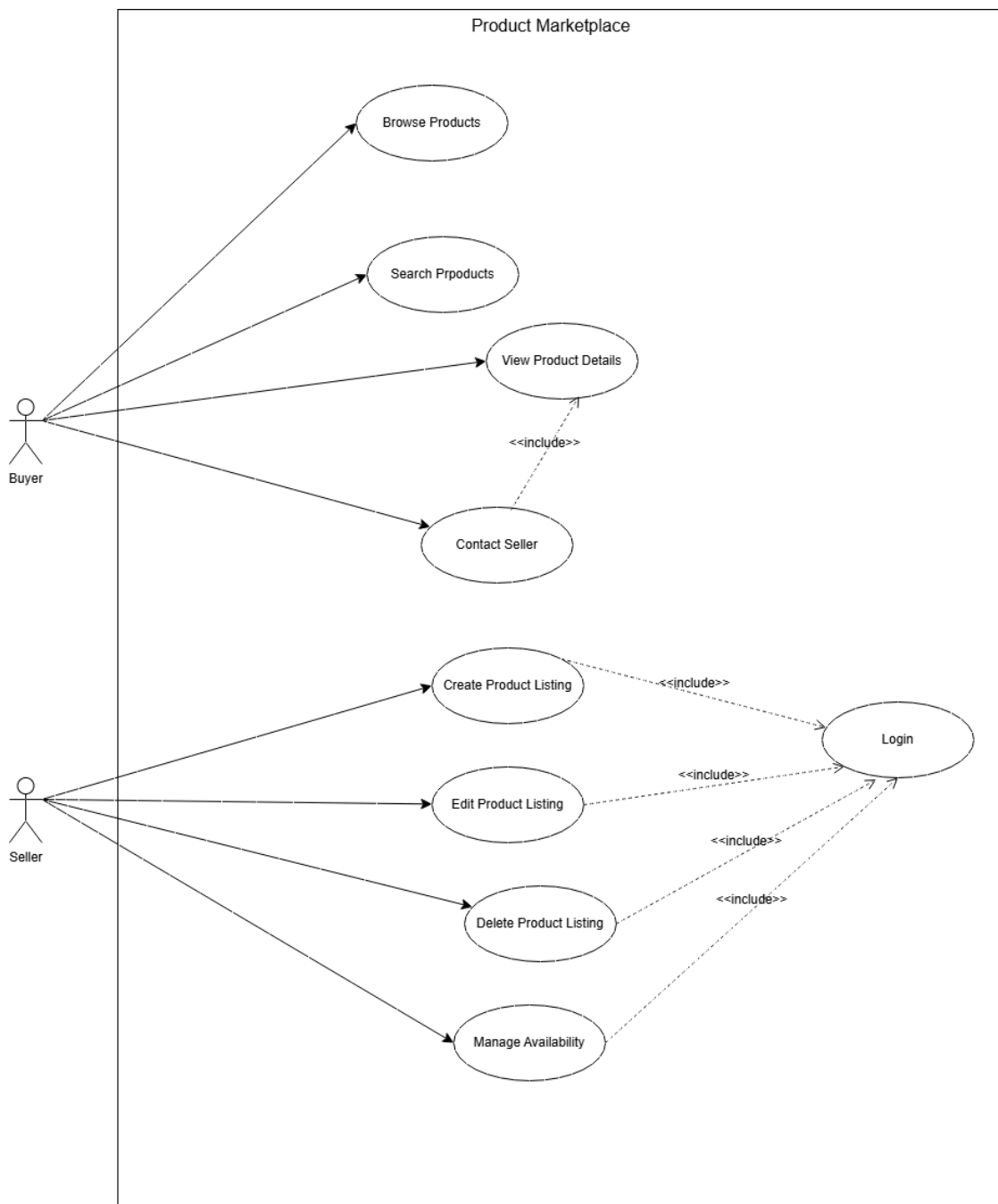
FR-06: The system shall assign permissions according to the user's role (Estate Owner, Estate Manager, Buyer, Seller, Worker/ Service Provider, or Administrator).

FR-07: The system shall allow users to securely log out.

3.2 Product Marketplace Module

Description

This module enables sellers to advertise coconut products while allowing buyers to search, view, and purchase products through the platform.



High-Level Description

Item	Description
Diagram Name	Product Marketplace
Purpose	Shows how sellers list products and buyers search, view, and contact sellers.
Primary Actors	Buyer, Seller
Related Functional Requirements	FR-06 Product Listing, FR-07 Product Search, FR-08 Contact Seller
Precondition	Seller is logged in to create listings. Buyers can browse available products.
Postcondition	Products are listed or buyers successfully contact sellers.

Functional Requirements

FR-08: Sellers shall be able to create product listings.

FR-09: Sellers shall be able to edit or remove existing product listings.

FR-10: Buyers shall be able to browse available products.

FR-11: Buyers shall be able to search and filter products by category, location, and price.

FR-12: Buyers shall be able to view detailed product information.

FR-13: Sellers shall be able to specify:

- Product category

- Selling method (Per Nut / Per Kilogram)
- Price
- Quantity
- Estate location
- Product images

FR-14: The system shall support selling:

- DC Coconut
- Curry Coconut
- Coconut Husk
- Other products like pepper,lemon,pineapple ,eggs
And other future products

FR-15: Buyers shall be able to contact sellers through the platform.

FR-16: Sellers shall be able to manage product availability.

3.3 Workforce Marketplace Module

Description

The Workforce Marketplace enables estate owners to find skilled workers and service providers while allowing workers to advertise their services.



High-Level Description

Item	Description
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Diagram Name	Workforce Marketplace
Purpose	Describes how estate owners hire workers and how workers or service providers manage job requests.
Primary Actors	Estate Owner, Worker, Service Provider
Related Functional Requirements	FR-09 Workforce Profile, FR-10 Hire Workers, FR-11 Job Request Management
Precondition	Service providers have registered their profiles.
Postcondition	Job requests are sent, accepted, rejected, or completed.

Functional Requirements

FR-17: Workers shall be able to create service profiles.

FR-18: Estate owners shall be able to search workers by service category.

FR-19: Workers shall be categorized under:

- Coconut Plucking
- Husking
- Fertilizing
- Coconut Gathering
- Coconut Loading
- Tractor Services
- Equipment Supply

FR-20: Workers shall be able to specify their charging method (Per Tree or Per Day).

FR-21: Estate owners shall be able to send job requests to workers.

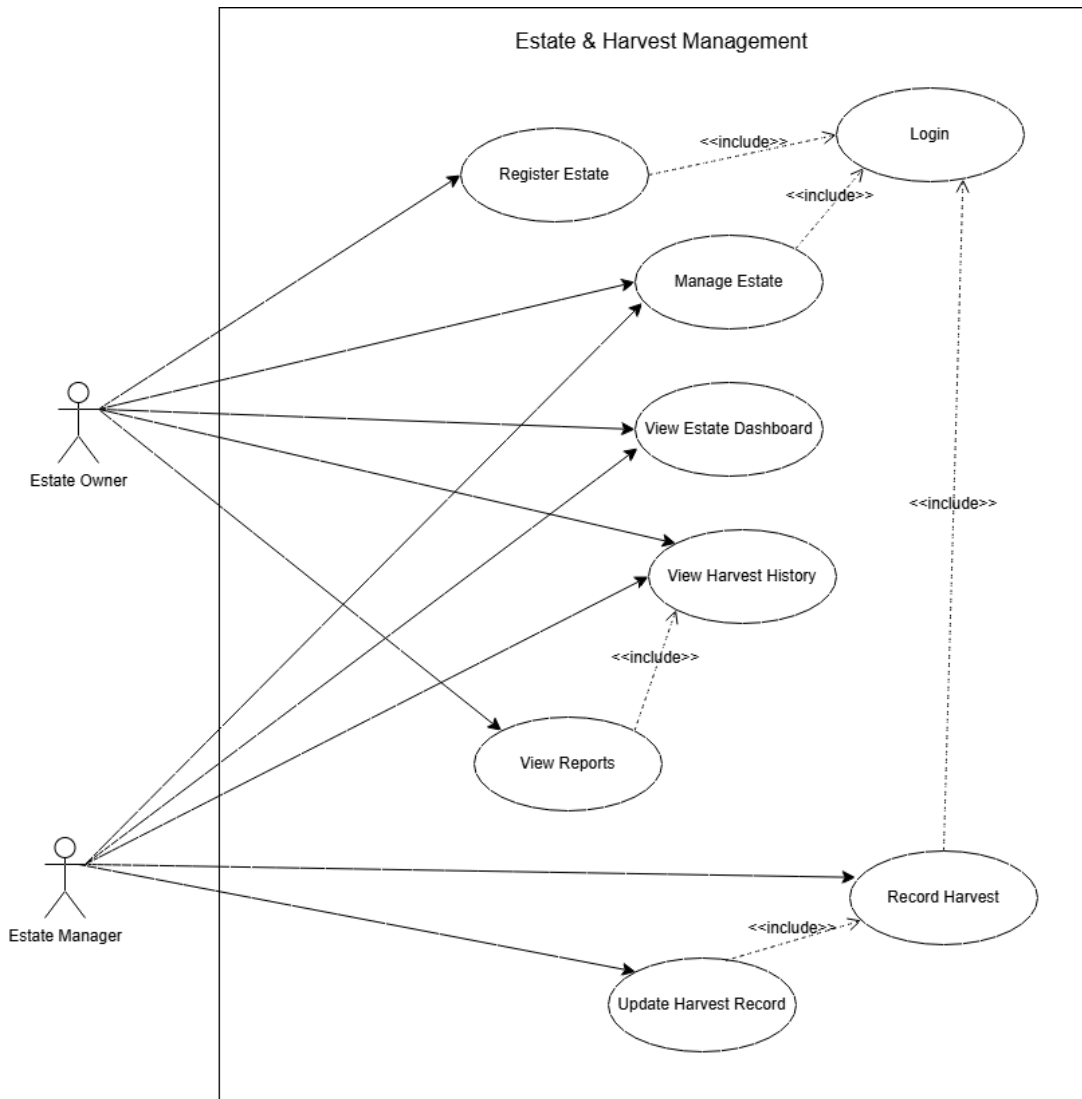
FR-22: Workers shall be able to accept or reject job requests.

FR-23: The system shall maintain the status of each job request.

3.4 Estate Management Module

Description

The Estate & Harvest Management module enables estate owners and estate managers to register estates, manage estate information, record harvests, and monitor harvest history and reports.



High-Level Description

Item	Description
Diagram Name	Estate & Harvest Management
Purpose	Illustrates estate registration, estate management, harvest recording, and harvest monitoring.
Primary Actors	Estate Owner, Estate Manager
Related Functional Requirements	FR-04 Estate Management, FR-12 Harvest Recording, FR-13 Harvest History
Precondition	The estate has been registered in the system.
Postcondition	Estate information and harvest records are updated successfully.

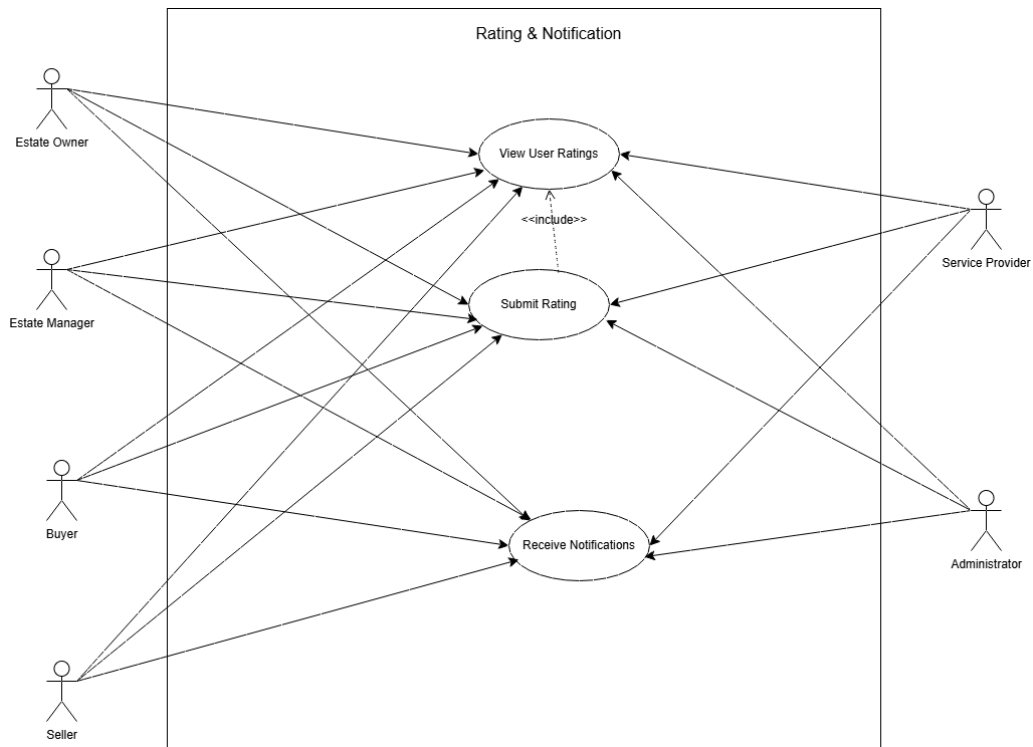
Functional Requirements

- FR-24 Register Estate
- FR-25 Update Estate
- FR-26 Store Estate Information
- FR-27 View Estate Statistics
- FR-28 Record Harvest
- FR-29 Store Harvest Details
- FR-30 View Harvest History
- FR-31 Generate Harvest Reports

3.5 Rating & Notification Module

Description

The Rating & Notification module allows users to submit ratings and reviews after completed transactions or jobs while receiving notifications about important platform activities.



High-Level Description

Item	Description
Diagram Name	Rating & Notification
Purpose	Demonstrates how users provide ratings after transactions and receive system notifications.
Primary Actors	Estate Owner, Buyer, Seller, Worker, Service Provider

Related Functional Requirements	FR-14 Rating & Reviews, FR-15 Notifications
Precondition	A transaction or job has been completed.
Postcondition	Ratings are recorded and notifications are delivered to users.

Functional Requirements

FR-32: Estate owners shall be able to rate workers.

FR-33: Workers shall be able to rate estate owners.

FR-34: Buyers shall be able to rate sellers.

FR-35: Sellers shall be able to rate buyers.

FR-36: Users shall be able to write reviews.

FR-37: The system shall calculate and display average ratings.

FR-38: The system shall notify users about new job requests.

FR-39: The system shall notify users when a job request is accepted or rejected.

FR-40: The system shall notify sellers when buyers contact them.

FR-41: The system shall notify users about important account activities.

3.8 Administration Module

Description

The Administration module enables administrators to monitor platform activities and maintain system integrity.



High-Level Description

Item	Description
Diagram Name	Administration
Purpose	Shows how administrators manage users, listings, reports, and platform activities.
Primary Actors	Administrator
Related Functional Requirements	FR-16 Administration
Precondition	Administrator is authenticated.

Postcondition	Administrative tasks are completed successfully.
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Functional Requirements

FR-42: Administrators shall verify newly registered users when required.

FR-43: Administrators shall manage user accounts.

FR-44: Administrators shall manage product listings.

FR-45: Administrators shall review reported users or inappropriate content.

FR-46: Administrators shall generate platform reports.

FR-47: Administrators shall suspend or remove users who violate platform policies.

4. External Interface Requirements

4.1 User Interface Requirements

The CocoConnect platform shall provide a user-friendly and responsive interface for both mobile and web users. The interface should be simple, consistent, and easy to navigate for users with different levels of technical knowledge.

The system shall provide separate interfaces based on user roles, including:

- Estate Owner Dashboard
- Estate Manager Dashboard
- Buyer Dashboard
- Seller Dashboard
- Worker Dashboard
- Service Provider Dashboard
- Administrator Dashboard

The user interface shall include:

- User Registration and Login pages
- Home Dashboard
- Product Marketplace
- Workforce Marketplace
- Estate Management
- Harvest Management
- Ratings and Reviews

- User Profile
- Notifications

4.2 Software Interface Requirements

The system shall interact with the following software components:

Software	Purpose
Flutter	Mobile and Web Frontend Development
Node.js	Backend Development
Express.js	REST API Development
PostgreSQL	Database Management
Firebase Cloud Messaging (Future)	Push Notifications
Google Maps API (Future)	Location and Estate Mapping

4.3 Hardware Interface Requirements

The application shall support:

- Android smartphones
- iPhones (iOS)
- Desktop and laptop computers
- Cloud-hosted database server

Minimum client requirements:

- Internet connection
- Modern web browser or mobile application
- Minimum 2 GB RAM
- Android 9.0 or above / iOS 14 or above

4.4 Communication Interface Requirements

Communication between the client application and the server shall use REST APIs over HTTPS.

The system shall support:

- Secure user authentication
- Data exchange using JSON format
- Internet-based communication
- Secure API requests and responses

5. Non-Functional Requirements

5.1 Performance

NFR-01: The system shall respond to user requests within 3 seconds under normal operating conditions.

NFR-02: The system shall support multiple users accessing the platform simultaneously without significant performance degradation.

NFR-03: Search results shall be displayed efficiently based on the user's search criteria.

5.2 Security

NFR-04: User passwords shall be securely encrypted before storage.

NFR-05: Only authenticated users shall access protected features.

NFR-06: The system shall implement role-based access control.

NFR-07: Personal information shall be protected from unauthorized access.

5.3 Reliability

NFR-08: The system shall provide reliable data storage with minimal risk of data loss.

NFR-09: User information and business records shall remain consistent after system failures.

5.4 Availability

NFR-10: The system should be available 24 hours a day, except during scheduled maintenance.

NFR-11: Planned maintenance activities should be communicated to users in advance whenever possible.

5.5 Usability

NFR-12: The application shall provide an intuitive interface that can be used with minimal training.

NFR-13: Navigation between modules shall be clear and consistent.

NFR-14: Error messages shall be simple and understandable.

5.6 Scalability

NFR-15: The system shall support future expansion to include additional agricultural products such as pepper, mango, pineapple, lemon, orange, poultry products, and tourism services.

NFR-16: The system architecture shall allow new modules to be integrated with minimal changes to the existing system.

5.7 Maintainability

NFR-17: The system shall be designed using a modular architecture to simplify maintenance.

NFR-18: Source code shall be documented to support future development and maintenance.

6. Business Rules

BR-01: Only registered users shall be allowed to access system services.

BR-02: A seller can publish only products that are currently available.

BR-03: Product listings shall include mandatory details such as product type, quantity, selling method, price, and location.

BR-04: Workers shall specify their service category and charging method (per tree or per day).

BR-05: Estate owners shall send job requests before assigning work to a worker.

BR-06: Workers may accept or reject job requests.

BR-07: Ratings and reviews shall only be submitted after a completed product transaction or completed service.

BR-08: Estate managers shall only record harvest information for estates assigned to them.

BR-09: Administrators have the authority to suspend user accounts that violate platform policies.

BR-10: Harvest records shall be maintained for future reporting and analysis.

7. Future Enhancements

The first version of CocoConnect focuses on supporting the coconut industry. Future releases may include the following enhancements:

- Online payment integration
- GPS-based worker and service provider location
- AI-based product recommendations
- AI-based workforce recommendations
- Weather forecasting for estate planning
- Inventory management
- Equipment rental management
- Marketplace for additional agricultural products such as pepper, mango, lemon, orange, pineapple, and poultry products
- Estate tourism and cabana booking
- Multilingual support
- Mobile push notifications
- Business analytics dashboard
- Integration with government agricultural services